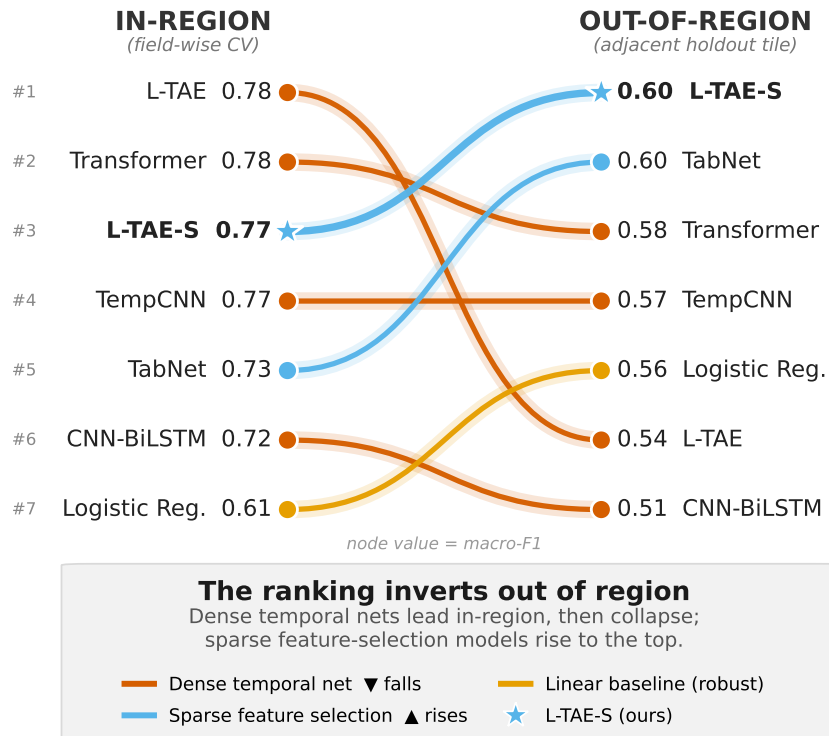


1 Graphical Abstract

2 **Sparse Feature Selection Outweighs Deep Learning Capacity for Out-**
3 **of-Region Transfer in Crop Classification**

4 Michael L. Mann, Sairam Venkatachalam, Disha Kacha, Devarsh Sheth, Ryan
5 Engstrom, Amir Jafari



Sparse Feature Selection Outweighs Deep Learning Capacity for Out-of-Region Transfer in Crop Classification

Michael L. Mann^{a,*}, Sairam Venkatachalam^b, Disha Kacha^b, Devarsh Sheth^b,
Ryan Engstrom^a, Amir Jafari^b

^a*Department of Geography & Environment, The George Washington
University, Washington, DC, 20052, USA*

^b*Data Science Program, The George Washington University, Washington, DC, 20052, USA*

Abstract

Crop-type maps from satellite image time series are a backbone of agricultural monitoring, yet the machine-learning models that produce them are almost always trained and tested within a single region, leaving open how they behave when deployed to new areas where no labels were collected. Using a South African Sentinel-2 dataset (optical only), we benchmark twenty-three classical machine-learning and deep-learning models under out-of-region transfer, applying them without any target-region labels or adaptation to an adjacent, non-overlapping holdout tile, and identify the model property most associated with success: sparse, axis-aligned feature selection. Models that commit to a few features at a time, gradient-boosted trees, logistic regression, and the attentive TabNet, transfer robustly (macro-F1 0.56–0.60), whereas high-capacity dense temporal networks (CNN-BiLSTM, a Transformer encoder, L-TAE) degrade sharply; in a controlled comparison, adding sparse feature selection to L-TAE improves transfer more than adding attention capacity does. We turn this insight into a method: L-TAE-S, a sparse-attention temporal encoder that ties the best model for out-of-region macro-F1 (0.60) at a fraction of the training cost, and whose field-level variant has the smallest train-to-holdout gap of any model. We further show that conventional in-region validation inverts these rankings, the dense networks that lead in-region (macro-F1 0.72–0.78) fall furthest under transfer, which is why models intended for operational crop mapping should be evaluated outside the region they were trained on. Field-level aggregation is a second, substitutable robustness lever, and the most transferable models retain near-full accuracy using only a quarter of the training fields. Together these results give practitioners a concrete design principle for crop classifiers that must be applied to nearby unlabeled farmland.

Keywords: Crop classification, deep learning, feature selection, remote sensing, Sentinel-2, out-of-region transfer, time-series classification

*Corresponding author.

Email address: mmann1123@email.gwu.edu (Michael L. Mann)
Preprint submitted to *Computers and Electronics in Agriculture*

1. Introduction

Accurate and timely crop classification is foundational to precision agriculture and to the agencies that monitor food security and agricultural land use, enabling yield forecasting, resource optimization, and food-security assessment (Bégué et al., 2020; Ienco et al., 2017). The Sentinel-2 constellation, with its multispectral richness and roughly five-day revisit, has made dense multi-temporal optical imagery widely available. Yet reliable crop classification remains hard throughout Africa and the Global South: spectral similarity between crops, irregular planting and phenological cycles, persistent cloud cover, and rain-fed dryland management all degrade accuracy (Bégué et al., 2020; Mahlayeye et al., 2024). Classical methods such as Random Forest (RF) and XGBoost have proven effective and efficient on engineered features (Mahlayeye et al., 2024; Mann et al., 2026; Mann and Warner, 2017; Mann et al., 2019), but their performance hinges on hand-crafted feature engineering (Maxwell et al., 2018; Belgiu and Drăguț, 2016). Deep learning, recurrent and convolutional temporal models, reduces reliance on hand-crafted predictors and captures phenological structure that summary statistics miss (Zhu et al., 2017; Rußwurm and Körner, 2018; Pelletier et al., 2019), but its generalizability in complex, low-label settings remains uncertain (Pankajakshan et al., 2025).

A second, and for our purposes critical, gap concerns *how generalization is measured*. Much of the crop-classification literature reports accuracy under random or field-wise cross-validation drawn from a single region (Belgiu and Drăguț, 2016; Garnot and Landrieu, 2020; Pelletier et al., 2019; Rußwurm and Körner, 2018; Ienco et al., 2017). Such protocols hold out individual pixels or fields but not *space*: training and test observations share the same atmosphere, soils, management calendars, and acquisition geometry, so the reported numbers reflect in-distribution fit rather than transfer to new terrain. Because operational maps are produced by applying a model to regions where no labels were collected, this distinction matters, yet studies that re-rank models under a spatially disjoint holdout remain rare (Pankajakshan et al., 2025; Jin et al., 2019; Rustowicz et al., 2019). This differs from the bulk of the transfer-learning literature, which adapts a model using target-domain data: labels, unlabeled imagery, or self-supervised pre-training (Ma et al., 2024; Wang et al., 2023; Mai et al., 2025). Our setting is the no-target-access case (domain generalization): models are trained on source tiles and applied to the holdout directly, with no adaptation, so what determines transfer is a property intrinsic to the model rather than a target-specific tuning step.

The Western Cape of South Africa concentrates the operational challenges, smaller fields, rain-fed dryland systems, heterogeneous planting rotations, and persistent winter cloud cover, while offering an unusually rich, high-quality labeled dataset (Radiant Earth Foundation, 2024). This makes it a realistic test bed: rich enough to train and compare modern architectures fairly, yet hard enough to expose the failure modes that matter in practice. We evaluate deep temporal models (CNN-BiLSTM, TabNet, L-TAE, a Transformer encoder, and TempCNN) and deep patch-based 2D/3D CNNs against classical XGBoost,

59 LightGBM, and RF. Critically, we evaluate every model twice: under field-wise
60 cross-validation *within* the training region, and on an adjacent, non-overlapping
61 holdout tile that no training pixel touches. We restrict inputs to multi-temporal
62 Sentinel-2 optical imagery alone, to isolate the spectral-time-series signal and
63 maximize applicability where only optical data are available. All code is public
64 (Mann, 2025a).

65 This study makes five contributions. First, a multi-architecture benchmark
66 (twenty-three classical, deep temporal, and deep patch-based models) showing
67 that in-region versus out-of-region transfer evaluation produce *inverted* rankings
68 on the same data. Second, empirical evidence that models with sparse, axis-
69 aligned feature-selection, the property behind two decades of RF and gradient-
70 boosted-tree dominance, are more stable when moved to a holdout tile, and
71 TabNet inherits it via sparsemax masks. Third, a controlled architectural compar-
72 ison, tested with paired bootstraps, in which adding sparse feature selection
73 improves transfer more than adding attention capacity, including L-TAE-S, a
74 sparse-attention encoder we introduce that ties TabNet for the best out-of-
75 region macro-F1 and achieves the smallest transfer gap in its field-level variant.
76 Fourth, identification of field-level aggregation as a second, substitutable robust-
77 ness lever that recovers transfer for dense models but is redundant for sparse
78 ones. Fifth, a data-efficiency analysis under out-of-region transfer showing the
79 most transferable models are also the most data-efficient, holding near-full hold-
80 out accuracy on a quarter of the training fields; as a supporting experiment, a
81 controlled comparison of `xr_fresh` time-series features against raw reflectance
82 for XGBoost and TabNet under out-of-region transfer. This leads to our central
83 question: across classical machine-learning and deep-learning models, which
84 most reliably classifies crops from multi-temporal Sentinel-2 optical imagery
85 when applied, without target-region labels, to an adjacent region outside the
86 training tiles, and what model properties determine whether in-distribution ac-
87 curacy survives the transfer?

88 1.1. Related Work

89 Early crop classification relied on classical learners (RF, SVM, logistic regres-
90 sion, gradient-boosted trees) operating on engineered descriptors (spectral in-
91 dices, temporal summaries) valued for efficiency, interpretability, and robustness
92 on structured data (Maxwell et al., 2018; Belgiu and Drăguț, 2016; Breiman,
93 2001; Chen and Guestrin, 2016; Ke et al., 2017), though their performance
94 depends on feature engineering that can limit transferability across agroeco-
95 logical regions (Wardlow et al., 2007; Mahlayeye et al., 2024). Deep models
96 learn representations directly from reflectance sequences: CNNs capture local
97 spectral-spatial pattern (Kussul et al., 2017), and recurrent and attention ar-
98 chitectures model seasonal phenology (Ienco et al., 2017; Rußwurm and Körner,
99 2018; Garnot and Landrieu, 2020; Rußwurm and Körner, 2020). Other designs
100 combine convolutional feature extraction with temporal modeling (Fan et al.,
101 2023; Tufail et al., 2025; Bandar and Coşkunçay, 2024), or use deep backbones as
102 feature extractors feeding classical classifiers (Espejo-Garcia et al., 2020; Koklu
103 et al., 2022; Yang et al., 2020). A recurring assumption is that richer learned

representations generalize better than compact engineered ones; whether this holds under out-of-region transfer, rather than within-region validation, is less established and is a central question here. Architectures that explicitly target cross-scenario generalization (e.g., Cropformer (Wang et al., 2023)) remain the exception and are rarely benchmarked against classical and deep baselines on a common spatially disjoint holdout. We fill this gap with a systematic comparison under *both* protocols, and additionally evaluate `xr_fresh` as an automated time-series feature-extraction framework in a crop-classification setting (Mann, 2025b).

2. Data

The study uses a labeled field dataset distributed through Radiant Earth’s MLHub (Radiant Earth Foundation, 2024) and used as the basis of the AI4FoodSecurity (South Africa) challenge (AI4EO, 2021), in an agriculturally productive part of the Western Cape (Fig. 1). Two adjacent Sentinel-2 tiles (34S_19E_258N, 34S_19E_259N) form the training region (4,150 labeled fields over 622 km²; mean field 15.0 ha); a third, immediately adjacent tile (34S_20E_259N) is reserved as a holdout for out-of-region transfer evaluation (2,417 fields over 282 km²). The tiles do not overlap, so no pixel or field is shared, and transfer must contend with different soils, geometries, and acquisition conditions. Because the holdout lies within the same agroecological zone and growing calendar, this is a relatively *local* shift, and the gaps we report are best read as a conservative lower bound on degradation under broader transfer. Throughout, “in-region” denotes a field-wise train/validation/test split within the training region; “out-of-region” (OOR) or “holdout” denotes predictions on the adjacent, non-overlapping tile, made without any target-region labels or adaptation. We use “out-of-region” rather than the broader “out-of-region transfer” because the holdout tests a local shift, not transfer across agroecological zones or years.

We classified five crop types: wheat, canola, lucerne/medics (alfalfa), barley, and small-grain grazing. The classes are heavily imbalanced; lucerne/medics is 43% of training fields and 56% of the holdout (Supplementary Material, Fig. S1(a)), which motivates macro-F1 as the primary metric (Sec. 3.3) and class-imbalance handling during training. Spectrally, wheat and barley overlap strongly, the dominant source of persistent confusion (Supplementary Material, Fig. S2(b)). Sentinel-2 Level-2A surface reflectance was acquired for the January–December 2017 growing season, cloud-masked with `s2cloudless` plus a shadow mask, and composited monthly. Six features per month are used: bands B2, B6, B11, B12, plus EVI and Hue, with EVI and Hue computed per scene before compositing, chosen to span the most informative spectral regions while limiting redundancy among correlated bands (Mann et al., 2026; Tufail et al., 2025). June 2017 yielded no usable imagery and May was discarded for cloud, leaving 10 monthly observations, so each pixel is a 60-dimensional time series (6 features $\times$ 10 months). Field polygons were digitized from sub-meter aerial photography and carry a single seasonal crop label (Radiant Earth Foundation, 2024); all predictions are made at the field level.

Study area: Western Cape, South Africa

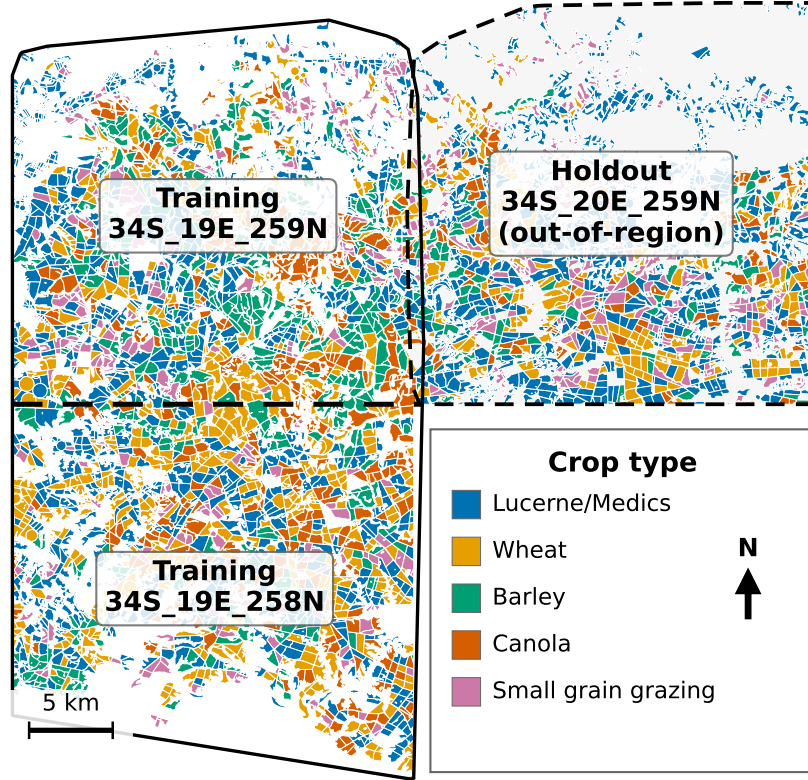


Figure 1: Study area in the Western Cape, South Africa. Two adjacent Sentinel-2 tiles (34S_19E_258N, 34S_19E_259N) form the training region; the immediately adjacent 34S_20E_259N tile (dashed) is held out for out-of-region transfer evaluation. Fields are colored by crop type; the tiles are spatially disjoint, so no pixel or field is shared between training and holdout.

3. Methodology

We compare two model families: classical machine learning on engineered `xr_fresh` features (9 models) and deep learning (14), the latter comprising deep temporal models on per-pixel or per-field time series (12) and deep patch-based models on image patches (2). To prevent leakage, all train/validation/test splitting is performed at the field level with a fixed random seed, so no field’s pixels straddle partitions; pixel- and patch-level predictions are aggregated to a field label by majority vote (or, for probabilistic models, mean pooling). Computational cost across paradigms is reported in the Supplementary Material (Sec. S-C).

3.1. Classical Machine Learning

We automate feature extraction with the `xr_fresh` toolkit, computing a set of statistical and temporal descriptors (e.g., autocorrelation, linear trend, day-of-year of the seasonal maximum, quantiles, complexity) from each pixel’s 10-month profile, yielding a 174-dimensional per-pixel vector (Mann et al., 2026; Mann, 2025b); the full feature list is in the Supplementary Material (Sec. S-A, Table S1). Four baselines: logistic regression, RF (300 trees), LightGBM, and XGBoost (James et al., 2013; Breiman, 2001; Ke et al., 2017; Chen and Guestrin, 2016; Friedman, 2001) are trained on this representation, all of them tree- or margin-based with built-in feature selection or shrinkage; standard formulations are reproduced in the Supplementary Material (Sec. S-A). We also report soft-voting and stacked ensembles (RF/XGBoost/LightGBM, with logistic-regression meta-learner), an Optuna-tuned field-level XGBoost, and a SMOTETomek-resampled variant; the *Stacking* and *SMOTE Stacked* pair differs only in resampling, isolating the effect of class rebalancing. At the field level, pixel features are mean-pooled within each polygon before classification (Peña-Barragán et al., 2011).

3.2. Deep Learning Architectures

At the pixel level each observation is a 60-feature temporal sequence. Although the pixel models train on millions of pixels, pixels within a field are highly autocorrelated and share one label, so the number of statistically independent units is closer to the number of fields; field-wise splitting keeps evaluation honest at the field scale. We evaluate, as five-seed ensembles, a CNN-BiLSTM trained with weighted focal loss; TabNet (Arik and Pfister, 2021; Muruganantham et al., 2024), which treats the observations as tabular input and, at each decision step, uses an attentive transformer to produce a *sparse* feature mask via sparsemax, selecting a handful of features and ignoring the rest; the L-TAE (Garnot and Landrieu, 2020), which applies a single learned query over the temporal sequence; a standard multi-head Transformer encoder (Vaswani et al., 2017; Rußwurm and Körner, 2020); and TempCNN (Pelletier et al., 2019). Detailed equations for these standard models, and the field-level voting rule, are in the Supplementary Material (Sec. S-A). The Transformer is included as an informative dense-attention comparison: it increases attention capacity relative to L-TAE while retaining no explicit sparse feature-selection mechanism. Deep patch-based models (a multi-channel 2D CNN and a 3D CNN) tile each field into 100 m patches resampled to 128×128 and are likewise trained as five-seed ensembles; details in the Supplementary Material (Sec. S-A).

3.2.1. Sparse-Attention L-TAE (L-TAE-S)

To test whether a tree-like, sparse inductive bias can be grafted onto a temporal-attention encoder, we introduce L-TAE-S, which inserts a TabNet-inspired channel-selection gate before the attention heads (Fig. 2). For each of the $n_{\text{head}} = 16$ heads, a per-head linear projection of the embedded, positionally-encoded sequence ($d_{\text{model}} = 128$, key dimension $d_k = 8$) is passed through

201 *sparsemax* (Martins and Astudillo, 2016) to produce a sparse, instance-dependent
 202 mask over embedding channels. Formally, for head h and time step $\mathbf{z}_t \in \mathbb{R}^E$,

$$\mathbf{m}_{h,t} = \text{sparsemax}(\mathbf{P}_{h,t} \odot \text{BN}(W_h \mathbf{z}_t)), \quad \tilde{\mathbf{z}}_{h,t} = \mathbf{m}_{h,t} \odot \mathbf{z}_t,$$

203

$$\mathbf{P}_{h+1,t} = \mathbf{P}_{h,t} \odot \max(\gamma - \mathbf{m}_{h,t}, 0), \quad \mathbf{P}_{1,t} = \mathbf{1};$$

204 head h then forms its keys and values from $\tilde{\mathbf{z}}_{h,t}$ before applying L-TAE attention.
 205 Because *sparsemax* projects onto the probability simplex, most channels of $\mathbf{m}_{h,t}$
 206 are exactly zero, and the relaxation prior $\mathbf{P}_{h,t}$ (coefficient $\gamma = 1.5$) discourages
 207 successive heads from re-selecting the same channels. These two mechanisms
 208 are the sole source of sparsity. No explicit sparsity penalty is used, because
 209 an ℓ_1 penalty on a sparsemax mask is constant ($\|\mathbf{m}_{h,t}\|_1 = 1$) and has zero
 210 gradient. This single block, the only architectural difference from the L-TAE,
 211 grafts a tree-like sparse feature-selection bias onto the attention backbone while
 212 keeping its temporal modeling, mirroring TabNet’s sparsemax masks (Arik and
 213 Pfister, 2021). We evaluate L-TAE-S at both the pixel and field level as a five-
 214 seed ensemble under the same weighted-focal objective and field-aggregation
 215 protocol as the other temporal networks.

216 3.3. Performance Metrics

217 Because the classes are severely imbalanced, plain accuracy is misleading.
 218 We adopt macro-F1 ($F1_m$, the unweighted mean of per-class F1) as the primary
 219 metric, since it weights every crop equally and penalizes failure on the minority
 220 cereals where models differ most. We report three further metrics: Cohen’s
 221 Kappa; weighted F1 (wF1, the support-weighted mean of per-class F1); and
 222 cross-entropy (Xent), the field-level scoring metric of the AI4FoodSecurity chal-
 223 lenge (AI4EO, 2021) from which the dataset is drawn (lower is better, computed
 224 on hard label predictions). Formal definitions are in the Supplementary Material
 225 (Sec. S-B).

226 3.4. Quantifying Uncertainty

227 We quantify two sources of variability. For sampling uncertainty in the finite
 228 holdout, we apply a nonparametric bootstrap over the 2,417 holdout fields: for
 229 each model we resample its per-field predictions with replacement $B = 10,000$
 230 times and recompute macro-F1, reporting the 2.5th–97.5th percentile interval
 231 as a 95% confidence interval (CI). For between-model differences, we use a
 232 paired bootstrap: each replicate resamples the holdout fields once and scores
 233 both models on the same fields, giving 95% CIs for Δ macro-F1, $\Delta\kappa$, and Δ
 234 accuracy; an exact McNemar test on per-field correctness serves as a check. For
 235 optimization noise, the per-seed standard deviation of OOR macro-F1 across the
 236 five ensemble members is 0.015–0.037 for the headline temporal models, which is
 237 why we report five-seed ensembles throughout. Both sources are small relative
 238 to the ranking effects below: every field-bootstrap CI spans only $\approx \pm 0.02$ –0.03
 239 macro-F1, so single-model intervals separate only for differences of ≈ 0.05 or
 240 more. Specific comparisons are therefore tested with the paired bootstrap, which
 241 is more sensitive because field difficulty cancels.

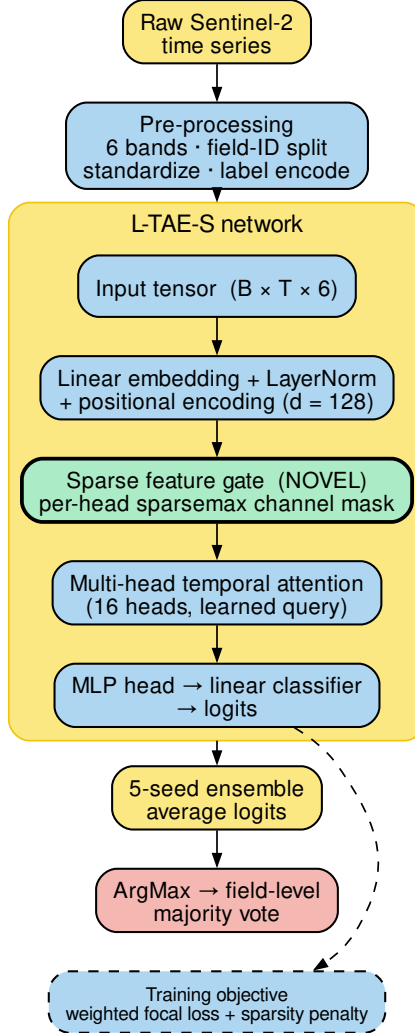


Figure 2: Architecture of the sparse-attention L-TAE-S. Before the multi-head temporal attention, a TabNet-style sparse feature gate (highlighted) applies a per-head **sparsemax** mask that selects a small subset of embedding channels, with a relaxation prior ($\gamma = 1.5$) discouraging heads from re-selecting the same channels. This block is the only architectural difference from the L-TAE; five seeds are ensembled and aggregated to field level.

4. Results

We evaluate every model under two protocols: 1) in-region field-wise validation and 2) out-of-region transfer (OOR) to the adjacent, non-overlapping holdout tile. All reported metrics are field-level. Pixel- and patch-level models are aggregated to field labels, whereas field-level models are trained and evaluated on field-aggregated inputs

4.1. In-Region Validation Overstates Performance and Misranks Models

Table 1 reports both protocols for all models, ranked by OOR macro-F1, with the gap $\Delta = F1_m^{\text{OOR}} - F1_m^{\text{in-region}}$. Two facts stand out. First, *every* model loses accuracy under transfer, but unevenly: from -0.04 (field-level L-TAE-S) to -0.24 (L-TAE at the pixel level). Second, this unevenness reorders the models. In-region the dense temporal pixel nets cluster at the top (L-TAE 0.78, Transformer 0.78, L-TAE-S 0.77, TempCNN 0.77, CNN-BiLSTM 0.72); a practitioner choosing by in-region F1 would pick from this cluster. On the holdout the cluster fractures: the Transformer drops to 0.58 and TempCNN to 0.57 ($\Delta = -0.20$), CNN-BiLSTM to 0.51 ($\Delta = -0.21$), and the in-region leader L-TAE ($\Delta = -0.24$) to 0.54, below every pixel-level tree learner.

The winners under transfer are a different group. The best single model is the TabNet pixel ensemble ($F1_m = 0.60$, $\kappa = 0.54$), which also posts the lowest holdout cross-entropy (4.42) of any model, tied on macro-F1 by both sparse-attention L-TAE-S variants (0.60). The most *robust* models, those with the smallest gaps, are the sparse learners: field-level L-TAE-S posts the single smallest gap ($\Delta = -0.04$), just ahead of logistic regression (-0.05) and the pixel-level tree learners. That L-TAE-S, a deep temporal attention network whose only twist is a learned gate restricting each head to a small channel subset, sits atop the robustness ranking is itself evidence for our thesis. The paired bootstrap confirms which differences are real: the best transferers beat the dense pixel nets by wide, significant margins (TabNet over CNN-BiLSTM, $\Delta = 0.09$, 95% CI $[0.07, 0.11]$; over L-TAE, $\Delta = 0.06$, $[0.04, 0.09]$), so the inversion is not holdout noise; at the very top the leaders (TabNet and both L-TAE-S variants) are statistically indistinguishable (all 0.60; TabNet vs. L-TAE-S, $p = 0.73$). We therefore do not claim a single best model; we claim that a sparse-feature-selection group occupies the top of the OOR ranking while the dense temporal nets prized by in-region validation fall significantly below it.

Table 1: In-region versus out-of-region transfer performance of all twenty-three models.

Model	Level	Features	In-reg		Out-of-region (OOR)				
			$F1_m$	Δ	$F1_m$	95% CI	κ	wF1	Xent
TabNet (raw)	pixel	raw	0.73	-0.13	0.60	.58-.63	0.54	0.70	4.42
L-TAE-S	field	raw	0.64	-0.04	0.60	.58-.62	0.52	0.69	4.77
L-TAE-S	pixel	raw	0.77	-0.17	0.60	.57-.62	0.53	0.70	4.63
L-TAE	field	raw	0.67	-0.07	0.59	.57-.62	0.47	0.67	5.51
Transformer	pixel	raw	0.78	-0.20	0.58	.55-.60	0.50	0.69	4.94
TempCNN	pixel	raw	0.77	-0.20	0.57	.55-.60	0.49	0.68	4.96

Model	Level	Features	F1 _m	Δ	F1 _m	95% CI	κ	wF1	Xent
XGBoost	field	xr_fresh	0.70	−0.13	0.56	.54–.59	0.47	0.68	4.89
LightGBM	pixel	xr_fresh	0.62	−0.06	0.56	.54–.59	0.47	0.66	4.85
Logistic Regression	pixel	xr_fresh	0.61	−0.05	0.56	.54–.59	0.47	0.67	4.93
LightGBM	field	xr_fresh	0.68	−0.12	0.56	.53–.58	0.45	0.66	4.99
Voting	pixel	xr_fresh	0.69	−0.13	0.56	.53–.58	0.48	0.66	4.85
Random Forest	pixel	xr_fresh	0.65	−0.10	0.55	.52–.57	0.45	0.65	5.07
XGBoost	pixel	xr_fresh	0.64	−0.09	0.54	.52–.57	0.47	0.66	5.01
L-TAE	pixel	raw	0.78	−0.24	0.54	.51–.56	0.40	0.62	6.42
TempCNN	field	raw	0.67	−0.14	0.54	.51–.56	0.43	0.64	6.07
Stacking	field	xr_fresh	0.64	−0.11	0.53	.51–.56	0.48	0.67	5.32
CNN-BiLSTM	pixel	raw	0.72	−0.21	0.51	.49–.54	0.39	0.61	6.69
CNN-BiLSTM	field	raw	0.64	−0.14	0.51	.48–.53	0.33	0.54	7.86
3D CNN	patch	raw	0.70	−0.20	0.50	.48–.53	0.29	0.54	7.98
SMOTE Stacked	field	xr_fresh	0.69	−0.21	0.49	.46–.51	0.43	0.63	5.37
Multi-Channel 2D CNN	patch	raw	0.63	−0.18	0.45	.42–.47	0.30	0.54	7.62
TabNet (raw, field-avg)	field	raw	0.62	−0.19	0.43	.41–.46	0.31	0.56	7.35
TabNet (xr_fresh)	field	xr_fresh	0.51	−0.15	0.37	.34–.40	0.25	0.53	6.13

Ranked by OOR macro-F1. In-region uses field-wise validation; OOR uses the disjoint holdout tile (34S_20E_259N). Δ is the gap; the 95% CI is the percentile bootstrap on OOR macro-F1 from 10,000 resamples of the 2,417 holdout fields. Xent is the AI4FoodSecurity field-level scoring metric (lower is better). *Features* marks the automated **xr_fresh** statistics (174 features) versus raw reflectance. Bold marks the best value per column.

4.2. Pixel Versus Field Level: Aggregation as Variance Reduction

Several architectures were run at both the pixel and the field level (averaging per-pixel sequences within a field before classification). Table 2 shows that field-level aggregation narrows the gap of every *dense* temporal network (L-TAE $-0.24 \rightarrow -0.07$, TempCNN $-0.20 \rightarrow -0.14$, CNN-BiLSTM $-0.21 \rightarrow -0.14$), but it raises holdout accuracy only for L-TAE (F1_m $0.54 \rightarrow 0.59$; paired $\Delta = 0.05$, 95% CI $[0.03, 0.08]$). For TempCNN ($0.57 \rightarrow 0.54$) and CNN-BiLSTM ($0.51 \rightarrow 0.51$) the gap closes mainly because in-region accuracy falls, so aggregation acts as a variance-reduction regularizer that trades in-region fit for stability rather than a reliable accuracy gain. The sparse-attention L-TAE-S follows the L-TAE pattern despite its channel gating: its gap shrinks from -0.17 to -0.04 while OOR F1_m holds at 0.60, so the response to aggregation appears to depend on both the temporal backbone and the input granularity; the sparse gate does not eliminate the benefit of field-level aggregation. TabNet is the informative exception: aggregating its raw inputs to the field level *hurts* (F1_m $0.60 \rightarrow 0.43$), because its robustness already derives from sparse feature selection and field-level training starves it of the per-pixel observation volume it exploits. The tree learners transfer well at either level, and their pixel→field change is essentially neutral under transfer (XGBoost $0.54 \rightarrow 0.56$). In short, aggregation stabilizes dense models but raises their transfer accuracy only for L-TAE; sparse, tree-like models do not need it.

4.3. Per-Class Performance and Persistent Confusions

Figure 3 breaks accuracy down crop by crop on the holdout for four representative models: the best transferer (TabNet), our sparse-attention L-TAE-S,

Table 2: Pixel- versus field-level out-of-region transfer.

Architecture	Pixel		Field		Field input
	$F1_m$	Δ	$F1_m$	Δ	
L-TAE-S	0.60	-0.17	0.60	-0.04	raw-avg
L-TAE	0.54	-0.24	0.59	-0.07	raw-avg
XGBoost	0.54	-0.09	0.56	-0.13	xr_fresh
LightGBM	0.56	-0.06	0.56	-0.12	xr_fresh
TempCNN	0.57	-0.20	0.54	-0.14	raw-avg
CNN-BiLSTM	0.51	-0.21	0.51	-0.14	raw-avg
TabNet	0.60	-0.13	0.43	-0.19	raw-avg

Architectures run at both granularities, ranked by field-level OOR $F1_m$. The *Field input* column gives the field-level input (**xr_fresh** statistics for the trees, field-averaged raw sequence for the deep models). Field aggregation narrows the gap for the dense temporal nets but raises their OOR $F1_m$ only for L-TAE, is roughly neutral for the tree learners, and degrades TabNet. Bold marks the best value per column.

its dense counterpart L-TAE, and CNN-BiLSTM (the weakest pixel-level temporal network on the holdout). The errors are highly uneven. TabNet and L-TAE-S classify the dominant lucerne/medics and the distinctive canola well ($F1$ 0.88 and 0.81; 0.87 and 0.76), L-TAE less so (0.77 and 0.67), but every model struggles with the cereals and small-grain grazing: even TabNet reaches only 0.55 on barley, 0.48 on wheat, and 0.30 on small-grain grazing. Adding the sparsity gate (L-TAE $\rightarrow$ L-TAE-S) on the same backbone improves every class, most on the easier ones (lucerne/medics $0.77 \rightarrow 0.87$, canola $0.67 \rightarrow 0.76$) and modestly on the cereals (barley $0.47 \rightarrow 0.50$, wheat $0.43 \rightarrow 0.47$, small-grain grazing $0.36 \rightarrow 0.40$). CNN-BiLSTM tracks L-TAE on most classes but is weakest on barley (0.33), so the dense networks lose accuracy on easy and hard classes alike, not only on minority classes. The wheat–barley confusion is consistent with their spectral overlap and appears across every model.

4.4. Data Efficiency Under Out-of-Region Transfer

The most transferable models are also the most data-efficient. We retrain a representative set on random 75%, 50%, and 25% subsets of the training fields and re-evaluate on the holdout; because a single subsample can be unrepresentative, we repeat each fraction over five independent draws and plot the mean (Fig. 4). Averaged this way, the deep temporal models are essentially flat from full data down to a quarter of the fields (L-TAE and L-TAE-S vary by under 0.02 macro- $F1$ across fractions), while the tree, linear, and TabNet baselines lose only ~ 0.03 – 0.05 —modest given a four-fold cut in training data. The per-fraction wiggles seen in any single split are within draw-to-draw sampling variance: L-TAE-S, whose sparse channel gate makes it the most sensitive to *which* fields are sampled, swings ± 0.06 macro- $F1$ at the 50% fraction (versus ≤ 0.02 for the robust learners), so the apparent 50% “dip” of any one draw is sampling noise rather than a data-efficiency effect. Holdout accuracy is therefore limited far more by out-of-region transfer than by training-set size, reinforcing

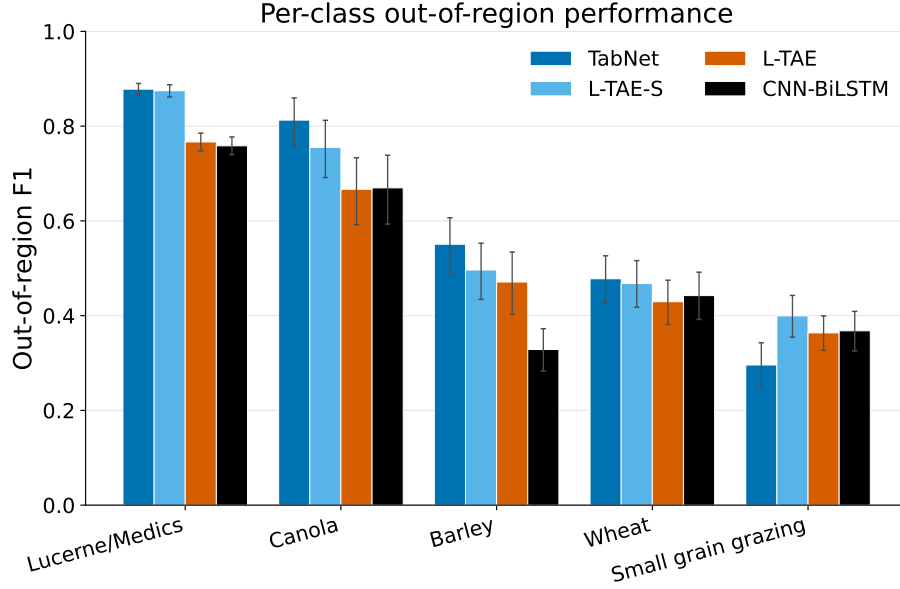


Figure 3: Per-class F1 on the holdout region for four representative models (TabNet, L-TAE-S, L-TAE, and CNN-BiLSTM). TabNet and L-TAE-S classify lucerne/medics and canola best; the cereals (barley, wheat) and small-grain grazing concentrate the errors. The dense L-TAE and CNN-BiLSTM also trail on the easier classes. Error bars are 95% bootstrap CIs over holdout fields.

333 that what governs holdout accuracy is the model’s inductive bias rather than
 334 the volume of training data.

335 4.5. Raw Reflectance versus *xr_fresh*

336 The best-transferring classical learners in Table 1 are exactly those fed
 337 *xr_fresh* statistics, so representation and model family are confounded there.
 338 To separate them, we trained a single XGBoost learner once on raw monthly
 339 reflectance and once on the per-pixel *xr_fresh* statistics, holding the FID-wise
 340 splits, preprocessing, and field-level mean pooling fixed (Table 3). The two rep-
 341 resentations are essentially indistinguishable under transfer ($F1_m = 0.59$ raw
 342 vs. 0.56 *xr_fresh*, a difference within the holdout bootstrap interval of Table 1;
 343 equal weighted-F1 and κ), and raw reflectance is, if anything, marginally ahead
 344 in-region as well. Field-level TabNet, run with the same configuration and
 345 seeds on both inputs, points the same way: field-averaged raw sequences trans-
 346 fer better than *xr_fresh* (OOR $F1_m = 0.43$ vs. 0.37 , non-overlapping bootstrap
 347 intervals; Table 1). For these two learners, *xr_fresh* yields no transfer-accuracy
 348 gain over raw reflectance. Its value is operational: a compact, fast-to-fit, agro-
 349 nomically interpretable vector that trains in seconds (Supplementary Material,
 350 Sec. S-C). The comparison does not extend to the deep temporal models, be-
 351 cause *xr_fresh* reduces each time series to summary statistics with no time

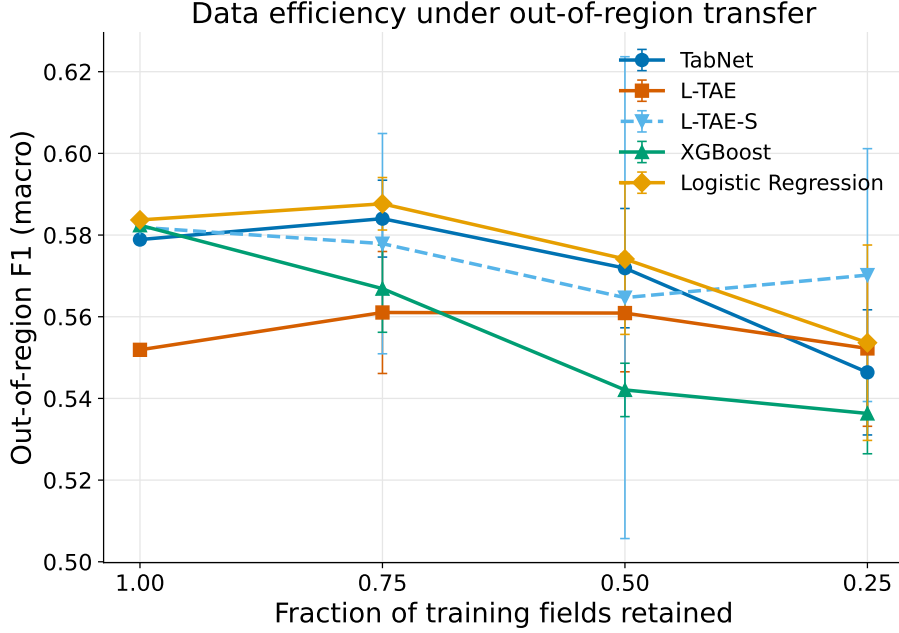


Figure 4: Data efficiency under out-of-region transfer: macro-F1 versus fraction of training fields retained, single training seed, each point the mean over five independent subsample draws (error bars ± 1 s.d.). The deep temporal models (L-TAE, L-TAE-S) are nearly flat to a quarter of the fields; the tree, linear, and TabNet baselines decline modestly. Per-fraction variation is within draw-to-draw sampling noise, largest for L-TAE-S (± 0.06 at 50%), so single-split “dips” are not real. Absolute values run ~ 0.02 – 0.05 below the five-seed levels of Table 1 because a single training seed is used.

Table 3: Raw reflectance versus `xr_fresh` for a single fixed XGBoost learner.

Representation	Split	$F1_m$	wF1	κ
Raw reflectance	In-region	0.73	0.80	0.71
<code>xr_fresh</code>	In-region	0.72	0.76	0.68
Raw reflectance	Out-of-region	0.59	0.69	0.50
<code>xr_fresh</code>	Out-of-region	0.56	0.69	0.50

Identical FID-wise splits, preprocessing, and field-level mean pooling; only the input representation differs. Metrics are field-level macro-F1 ($F1_m$), weighted F1 (wF1), and Cohen’s κ .

axis for L-TAE, TempCNN, or CNN-BiLSTM to attend or convolve over. This is, to our knowledge, the first controlled comparison of `xr_fresh` against raw reflectance under out-of-region transfer.

355 5. Discussion

356 The conclusion about which model is “best” depends almost entirely on
 357 whether evaluation is in-region or under out-of-region transfer, and a model’s
 358 transfer behavior is predictable from its inductive bias.

359 5.1. Why Tree-Like Inductive Bias Transfers

360 The clearest pattern is that transfer is strongly associated with inductive
 361 bias, and not simply with model family or attention capacity. Figure 5 groups
 362 every model by inductive bias and plots its OOR gap: linear models lose the least
 363 (mean $\Delta = -0.05$), tree-based models little (mean -0.11), the sparse-attention
 364 L-TAE-S sits essentially with the trees (mean -0.11 across its variants), while
 365 dense temporal/patch nets lose the most (mean -0.20) and SMOTE-augmented
 366 stacking is similarly fragile (-0.21). RF and gradient-boosted trees have been
 367 the remote-sensing workhorse for two decades not because they top in-region
 368 accuracy (dense nets often do) but because they survive when a model is applied
 369 to an unseen region (the operational case) (Belgiu and Drăguț, 2016; Maxwell
 370 et al., 2018). A tree decides one feature at a time, choosing the few features that
 371 matter for each split and ignoring the rest; this sparsity may reduce reliance on
 372 region-specific quirks of phenology, soil, and atmosphere that pull richer models
 373 off course. Independent corroboration spans settings: a 3D-CNN collapses under
 374 cross-sensor hyperspectral transfer while a sparse SVM holds (Pankajakshan
 375 et al., 2025); RF matches or exceeds a dense CNN-LSTM on label-scarce Ghana
 376 fields (Rustowicz et al., 2019); a maize model drops 16 points moving across East
 377 Africa (Jin et al., 2019); and bagged/boosted trees out-transfer an elaborate
 378 instance-reweighting scheme from the U.S. to northwest China (Mai et al., 2025).
 379 TabNet, the best transferer here, is a neural network built to behave like a
 380 tree: a small learned gate picks a handful of features at each step (Arik and
 381 Pfister, 2021), so it clusters with the trees and inherits their transferability. By
 382 contrast, CNN-BiLSTM, TempCNN, L-TAE, and the Transformer consume the
 383 whole temporal sequence at once with no feature selection. At the pixel level
 384 this leaves them fitting the training region tightly: they post the largest in-
 385 region-to-holdout gaps in the benchmark, with L-TAE dropping furthest ($\Delta =$
 386 -0.24). This is consistent with evidence that the dates such models attend to
 387 are themselves contingent on the training region (Obadić et al., 2022). Dense
 388 attention is not intrinsically poor under out-of-region transfer: the pixel-level
 389 L-TAE loses 0.24 macro-F1, but the field-level L-TAE reduces this gap to 0.07
 390 and raises holdout macro-F1 to 0.59 (Sec. 4.2), so it is the pixel-level dense
 391 models specifically that transfer worst.

392 The Transformer and L-TAE-S together separate attention capacity from
 393 feature selection on the same pixel-level task. The Transformer deploys far
 394 richer attention than L-TAE’s single query and does transfer better ($F1_m$ 0.58
 395 vs. 0.54; paired $\Delta = 0.04$, 95% CI $[0.02, 0.06]$), so capacity is not irrelevant.
 396 Adding sparse feature selection to the smaller L-TAE helps more: L-TAE-S
 397 reaches 0.60 ($\Delta = 0.06$, $[0.04, 0.08]$; $\Delta\kappa = 0.13$) and matches or exceeds the

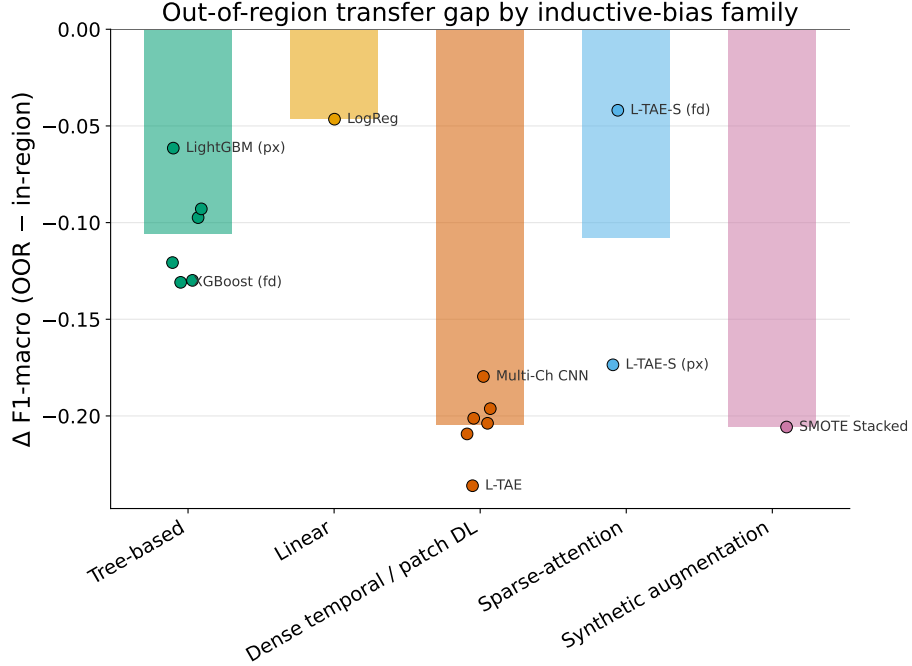


Figure 5: Out-of-region transfer gap (Δ macro-F1) grouped by inductive-bias family. Sparse/tree-like and linear models transfer (small $|\Delta|$); dense temporal/patch nets and synthetic augmentation overfit the training region. L-TAE-S, which adds sparse feature selection to a dense attention backbone, tracks the tree-based models. Bars are family means; points are individual models.

Transformer (Δ macro-F1 0.02, $p = 0.06$; $\Delta\kappa$ and Δ accuracy 0.02, both significant). At the field level the gate’s macro-F1 gain over L-TAE is not significant (0.60 vs. 0.59), but κ and accuracy improve ($\Delta = 0.04$ and 0.05, both significant). L-TAE-S and TabNet, the two models that explicitly select features before computing with them, are the best transferers. Within the architectures tested, sparse feature selection is therefore associated with larger and more consistent transfer gains than added attention capacity; this is consistent with, but does not prove, feature selection as the property that matters most. The L-TAE-S experiment also shows that this is *not* a tree-versus-deep tradeoff: the same selection step used by tree splits can be grafted onto a dense architecture at the cost of a single block, and L-TAE-S ties TabNet for the best holdout accuracy. That parity is not accuracy alone: at the pixel level L-TAE-S reaches TabNet’s best-in-class holdout accuracy in roughly 40% of the per-seed training time (24 vs. 61 min), and its field-level variant is both faster and more accurate than any field-aggregated TabNet (Supplementary Material, Sec. S-C). The next generation of operational crop classifiers need not choose between the expressivity of deep learning and the transfer-robustness of trees.

5.2. Two Robustness Levers

Two caveats sharpen rather than weaken the story, and together they point to how we can choose the best models. First, it is *pixel-level* TabNet that wins; the field-aggregated TabNet variants are among the worst transferers (Table 1), because aggregation removes the per-pixel observation volume the sparse model exploits. Second, dense temporal attention transfers well *when* applied at the field level: L-TAE’s pixel-level gap shrinks from -0.24 to -0.07 under aggregation, and L-TAE-S ends with an even smaller gap (Table 2); for TempCNN and CNN-BiLSTM aggregation narrows the gap without raising holdout accuracy. These point to two distinct, partly substitutable directions. The first lever is built-in feature selection. The model itself picking a few features at a time (trees, TabNet, and now L-TAE-S); this is intrinsic robustness paid for at design time. The second is variance reduction imposed from outside via field-level aggregation, which averages away region-specific pixel noise before the classifier sees it. Which provides external robustness available even when the architecture is fixed. A model with the first lever gains nothing, and may lose observation volume, from the second; this is exactly why TabNet prefers pixels. These two properties can be combined. L-TAE-S adds a sparse gate to a temporal-attention backbone and then benefits from field aggregation, ending with the smallest transfer gap in the benchmark. Practitioners can read this as a flowchart: if free to choose or modify the model, build feature selection in (lever 1); if constrained to an off-the-shelf dense architecture, aggregate to the field level (lever 2), which stabilizes transfer even when it does not raise accuracy. Two further controlled comparisons confirm the ranking is not an artifact of ensembling or imbalance handling; both shift absolute scores but never reorder the robust and fragile groups, with the full tables in the Supplementary Material (Sec. S-D). Seed-ensembling gives only a small, monotone lift that leaves the ordering intact. A single-seed L-TAE-S ($F1_m = 0.56$) already exceeds the five-seed CNN-BiLSTM and TabNet field ensembles (Table S4). And the imbalance-handling objective moves absolute scores but not the ranking: the robust L-TAE-S is almost invariant across five loss and sampler settings ($F1_m = 0.59\text{--}0.60$), whereas the fragile CNN-BiLSTM swings widely ($0.41\text{--}0.56$) and even at its best stays below L-TAE-S’s worst (Table S5).

5.3. Automated Features, Patch Models, and Persistent Difficulty

The classical pipelines paired with `xr_fresh` were not only robust but cheap, transferring as well as the best deep models (OOR $F1_m$ $0.55\text{--}0.56$) while training in minutes rather than hours (Supplementary Material, Sec. S-C); the controlled comparisons for XGBoost and TabNet (Sec. 4.5) show that this value is operational rather than an accuracy edge over raw reflectance. Patch-based convolutional models did not exceed pixel-level models and transferred poorly (3D CNN $F1_m = 0.50$; multi-channel 2D CNN 0.45). We tested one configuration of each, so this should not be read as a property of patch CNNs in general. Several design factors may also contribute to it. At 10 m resolution a 100 m patch holds only $\sim 10 \times 10$ pixels, and upsampling to 128×128 adds interpolation rather

than detail. After upsampling, the convolutional receptive fields span only one to two native pixels before global pooling or flattening, so little spatial structure is learned. The two models also fuse time differently: the 2D CNN stacks all band-month images as input channels, discarding temporal order, while the 3D CNN convolves across months. Lack of sparse feature selection may also play a role. Our experiments do not separate these factors; patch-size and architecture sweeps would be needed to do so. Finally, small-grain grazing, barley, and wheat are the hardest classes for every model (Fig. 3) and account for most accuracy lost under transfer; that even robust models struggle here, and that the dense networks trail on the easier classes as well, indicates the limitation is partly intrinsic spectral similarity and partly the transfer problem itself, not class imbalance alone.

5.4. Limitations

The study uses a single growing season (2017) and a single holdout tile that is spatially *adjacent* to the training tiles within the same agroecological zone. This is a relatively local shift, so the gaps we report most likely *understate* the degradation that fully cross-regional or cross-year transfer would produce; our numbers are best read as a conservative lower bound, and multi-region, multi-year holdouts are needed to confirm the inductive-bias ordering is stable under larger shifts. Our conclusions should therefore be read as applying to out-of-region transfer between neighboring tiles, not to cross-ecozone or cross-year transfer. The labels derive from a public dataset rather than our own campaign, and we use Sentinel-2 optical data only; a deliberate scoping choice that nonetheless limits the achievable ceiling for optically similar cereals, making SAR fusion a clear next step.

6. Conclusion

In-region validation, the protocol the crop-classification literature reports almost universally, misranks twenty-three Sentinel-2 classifiers when they are applied to an adjacent, non-overlapping holdout tile. Dense temporal networks that lead under conventional evaluation fall sharply under transfer, the in-region leader (L-TAE) dropping below every pixel-level tree learner, while parsimonious models such as tree ensembles, logistic regression, and a TabNet ensemble retain rank and accuracy. The common thread is the same property that made tree ensembles the remote-sensing default for two decades: a sparse, axis-aligned feature-selection bias that commits to a few robust signals and ignores the rest. TabNet inherits it through sparsemax masks. A Transformer with far greater attention capacity transfers better than the lightweight L-TAE, but adding a sparse gate to L-TAE helps more, pointing to sparsity rather than capacity as the stronger lever. Grafting that lever onto an attention model (L-TAE-S) ties TabNet for the best transfer at a fraction of its training cost and, at the field level, posts the smallest in-region-to-holdout gap of any model. Field-level aggregation is a second, partly substitutable lever: it narrows the transfer gap

of dense temporal networks, raising holdout accuracy for L-TAE and L-TAE-S, but hurts TabNet. We therefore recommend the following: evaluate on an out-of-region holdout before declaring one architecture better than another; prefer models with a built-in feature-selection bias when mapping nearby unlabeled areas; and, if dense temporal models are required, aggregate their predictions to the field level. The conditions that make the Western Cape difficult are the conditions under which agricultural remote sensing must operate across much of the Global South, and evaluating models the way they are actually deployed, applied to regions where no labels were collected, is what will make remote sensing deliver where it matters most.

CRedit authorship contribution statement

Michael L. Mann: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Sairam Venkatachalam:** Software, Formal analysis, Writing – original draft, Investigation. **Disha Kacha:** Software, Formal analysis, Investigation, Writing – original draft. **Devarsh Sheth:** Software, Formal analysis, Investigation, Writing – original draft. **Ryan Engstrom:** Supervision, Methodology, Writing – review & editing. **Amir Jafari:** Methodology, Methodology, Supervision, Writing – review & editing.

Ethics statement

This article does not contain any studies with human participants or animals performed by any of the authors. The study is based solely on publicly available satellite imagery (Sentinel-2) and field-boundary crop labels, so no ethical approval was required.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The processed dataset supporting this study is openly available on IEEE DataPort at <https://dx.doi.org/10.21227/n69k-r458> (DOI: 10.21227/n69k-r458). The source code is available at https://github.com/mmenn1123/South_Africa_Crop_Comp (Mann, 2025a). The study builds on the ground-reference crop labels of the Western Cape Department of Agriculture and Radiant Earth Foundation, “Crop Type Classification Dataset for Western Cape, South Africa”, v1.0, Radiant MLHub (DOI: 10.34911/rdnt.j0co8q), used under the CC-BY-4.0 license; Sentinel-2 imagery was extracted via Google Earth Engine.

538 Acknowledgements

539 This research did not receive any specific grant from funding agencies in
540 the public, commercial, or not-for-profit sectors. We thank the Western Cape
541 Department of Agriculture and the Radiant Earth Foundation for collecting and
542 openly distributing the labeled field dataset.

543 References

- 544 AI4EO, 2021. AI4FoodSecurity challenge (South Africa). European Space
545 Agency (ESA), in partnership with Planet, TUM, DLR and Radiant Earth,
546 <https://platform.ai4eo.eu/ai4food-security-south-africa>.
- 547 Arik, S.Ö., Pfister, T., 2021. TabNet: Attentive interpretable tabular learning.
548 Proceedings of the AAAI Conference on Artificial Intelligence 35, 6679–6687.
549 doi:10.1609/aaai.v35i8.16826.
- 550 Bandar, A., Coşkunçay, A., 2024. Crop classification with attention based
551 BI-LSTM and temporal convolution neural network combination for remote
552 sensing BreizhCrop time series data. *Yüzüncü Yıl Üniversitesi Fen Bilimleri*
553 *Enstitüsü Dergisi* 29, 173–188. doi:10.53433/yyufbed.1335866.
- 554 Bégué, A., Leroux, L., Soumaré, M., Faure, J.F., Diouf, A.A., Augusseau, X.,
555 Touré, L., Tonneau, J.P., 2020. Remote sensing products and services in sup-
556 port of agricultural public policies in Africa: overview and challenges. *Front-*
557 *iers in Sustainable Food Systems* 4, 58. doi:10.3389/fsufs.2020.00058.
- 558 Belgiu, M., Drăguţ, L., 2016. Random forest in remote sensing: A review of
559 applications and future directions. *ISPRS Journal of Photogrammetry and*
560 *Remote Sensing* 114, 24–31. doi:10.1016/j.isprsjprs.2016.01.011.
- 561 Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.1023/A:
562 1010933404324.
- 563 Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system,
564 in: Proceedings of the 22nd ACM SIGKDD International Conference on
565 Knowledge Discovery and Data Mining, pp. 785–794. doi:10.1145/2939672.
566 2939785.
- 567 Espejo-Garcia, B., Mylonas, N., Athanasakos, L., Fountas, S., Vasilakoglou, I.,
568 2020. Towards weeds identification assistance through transfer learning. *Com-*
569 *puters and Electronics in Agriculture* 171, 105306. doi:10.1016/j.compag.
570 2020.105306.
- 571 Fan, X., Chen, L., Xu, X., Yan, C., Fan, J., Li, X., 2023. Land cover classification
572 of remote sensing images based on hierarchical convolutional recurrent neural
573 network. *Forests* 14, 1881. doi:10.3390/f14091881.

574 Friedman, J.H., 2001. Greedy function approximation: A gradient boosting
575 machine. *Annals of Statistics* 29, 1189–1232. doi:10.1214/aos/1013203451.

576 Garnot, V.S.F., Landrieu, L., 2020. Lightweight temporal self-attention for
577 classifying satellite images time series, in: *Advanced Analytics and Learning
578 on Temporal Data (AALTD 2020)*, ECML PKDD Workshops, Springer. pp.
579 171–181. doi:10.1007/978-3-030-65742-0_12.

580 Ienco, D., Gaetano, R., Dupaquier, C., Maurel, P., 2017. Land cover classifi-
581 cation via multitemporal spatial data by recurrent neural networks. *arXiv
582 preprint arXiv:1704.04055* .

583 James, G., Witten, D., Hastie, T., Tibshirani, R., 2013. *An Introduction to
584 Statistical Learning*. Springer.

585 Jin, Z., Azzari, G., You, C., Di Tommaso, S., Aston, S., Burke, M., Lobell,
586 D.B., 2019. Smallholder maize area and yield mapping at national scales
587 with Google Earth Engine. *Remote Sensing of Environment* 228, 115–128.
588 doi:10.1016/j.rse.2019.04.016.

589 Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu,
590 T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree, in:
591 *Advances in Neural Information Processing Systems (NeurIPS)*.

592 Koklu, M., Unlarsen, M.F., Ozkan, I.A., Aslan, M.F., Sabanci, K., 2022. A
593 CNN-SVM study based on selected deep features for grapevine leaves clas-
594 sification. *Measurement* 188, 110425. doi:10.1016/j.measurement.2021.
595 110425.

596 Kussul, N., Lavreniuk, M., Skakun, S., Shelestov, A., 2017. Deep learning
597 classification of land cover and crop types using remote sensing data. *IEEE
598 Geoscience and Remote Sensing Letters* 14, 778–782. doi:10.1109/LGRS.
599 2017.2681128.

600 Ma, Y., Chen, S., Ermon, S., Lobell, D.B., 2024. Transfer learning in en-
601 vironmental remote sensing. *Remote Sensing of Environment* 301, 113924.
602 doi:10.1016/j.rse.2023.113924.

603 Mahlayeye, M., Darvishzadeh, R., Nelson, A., 2024. Characterising maize and
604 intercropped maize spectral signatures for cropping pattern classification. *In-
605 ternational Journal of Applied Earth Observation and Geoinformation* 128,
606 103699. doi:10.1016/j.jag.2024.103699.

607 Mai, J., Feng, Q., Fu, S., Wang, R., Zhang, S., Zhang, R., Liang, T.,
608 2025. Enhancing crop type mapping in data-scarce regions through trans-
609 fer learning: A case study of the Hexi Corridor. *Remote Sensing* 17, 1494.
610 doi:10.3390/rs17091494.

611 Mann, M.L., 2025a. South_Africa_Crop_Comp: Source code for multitempo-
612 ral crop classification. GitHub repository, [https://github.com/mmnn1123/](https://github.com/mmnn1123/South_Africa_Crop_Comp)
613 [South_Africa_Crop_Comp](https://github.com/mmnn1123/South_Africa_Crop_Comp).

614 Mann, M.L., 2025b. xr_fresh: Automated time series feature extraction for
615 remote sensing and gridded data. *Journal of Open Source Software* 10, 9009.
616 doi:10.21105/joss.09009.

617 Mann, M.L., Colson, L., Nealon, R., Engstrom, R., Nakacwa, S., 2026. Lite
618 learning: Efficient crop classification in Tanzania using feature extraction
619 with machine learning and crowd sourcing. *IEEE Journal of Selected Topics*
620 *in Applied Earth Observations and Remote Sensing* In press.

621 Mann, M.L., Warner, J.M., 2017. Ethiopian wheat yield and yield gap estima-
622 tion: A spatially explicit small area integrated data approach. *Field Crops*
623 *Research* 201, 60–74. doi:10.1016/j.fcr.2016.10.014.

624 Mann, M.L., Warner, J.M., Malik, A.S., 2019. Predicting high-magnitude, low-
625 frequency crop losses using machine learning: an application to cereal crops in
626 Ethiopia. *Climatic Change* 154, 211–227. doi:10.1007/s10584-019-02432-7.

627 Martins, A.F.T., Astudillo, R.F., 2016. From softmax to sparsemax: A sparse
628 model of attention and multi-label classification, in: *Proceedings of the 33rd*
629 *International Conference on Machine Learning (ICML)*, pp. 1614–1623.

630 Maxwell, A.E., Warner, T.A., Fang, F., 2018. Implementation of machine-
631 learning classification in remote sensing: an applied review. *International*
632 *Journal of Remote Sensing* 39, 2784–2817. doi:10.1080/01431161.2018.
633 1433343.

634 Muruganantham, P., Wibowo, S., Grandhi, S., Islam, N., 2024. A deep learning
635 approach for dealing with tabular data in crop classification, in: *Proceedings*
636 *of the IEEE International Conference on ICT Convergence (ICTC)*, pp. 2054–
637 2059. doi:10.1109/ICTC62082.2024.10826760.

638 Obadic, I., Roscher, R., Oliveira, D.A.B., Zhu, X.X., 2022. Exploring
639 self-attention for crop-type classification explainability. *arXiv preprint*
640 *arXiv:2210.13167*.

641 Pankajakshan, P., Padmasanan, A., Sundar, S., 2025. Deep architectures fail
642 to generalize: A lightweight alternative for agricultural domain transfer in
643 hyperspectral images. *Sensors* 26, 174. doi:10.3390/s26010174.

644 Pelletier, C., Webb, G.I., Petitjean, F., 2019. Temporal convolutional neural
645 network for the classification of satellite image time series. *Remote Sensing*
646 11, 523. doi:10.3390/rs11050523.

647 Peña-Barragán, J.M., Ngugi, M.K., Plant, R.E., Six, J., 2011. Object-based
648 crop identification using multiple vegetation indices, textural features and
649 crop phenology. *Remote Sensing of Environment* 115, 1301–1316. doi:10.
650 1016/j.rse.2011.01.009.

651 Radiant Earth Foundation, 2024. South Africa crop type competition. Source
652 Cooperative repository. Data DOI: 10.34911/rdnt.j0co8q.

653 Rußwurm, M., Körner, M., 2018. Multi-temporal land cover classification
654 with sequential recurrent encoders. *ISPRS International Journal of Geo-*
655 *Information* 7, 129. doi:10.3390/ijgi7040129.

656 Rußwurm, M., Körner, M., 2020. Self-attention for raw optical satellite time
657 series classification. *ISPRS Journal of Photogrammetry and Remote Sensing*
658 169, 421–435. doi:10.1016/j.isprsjprs.2020.06.006.

659 Rustowicz, R.M., Cheong, R., Wang, L., Ermon, S., Burke, M., Lobell, D., 2019.
660 Semantic segmentation of crop type in Africa: A novel dataset and analysis
661 of deep learning methods, in: *Proceedings of the IEEE/CVF Conference on*
662 *Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 75–82.

663 Tufail, R., Tassinari, P., Torreggiani, D., 2025. Deep learning applications for
664 crop mapping using multi-temporal Sentinel-2 data and red-edge vegetation
665 indices: Integrating convolutional and recurrent neural networks. *Remote*
666 *Sensing* 17, 3207. doi:10.3390/rs17183207.

667 Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N.,
668 Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: *Advances in*
669 *Neural Information Processing Systems (NeurIPS)*, pp. 5998–6008.

670 Wang, H., Chang, W., Yao, Y., Yao, Z., Zhao, Y., Li, S., Liu, Z., Zhang, X.,
671 2023. Cropformer: A new generalized deep learning classification approach
672 for multi-scenario crop classification. *Frontiers in Plant Science* 14, 1130659.
673 doi:10.3389/fpls.2023.1130659.

674 Wardlow, B.D., Egbert, S.L., Kastens, J.H., 2007. Analysis of time-series
675 MODIS 250 m vegetation index data for crop classification in the U.S. Central
676 Great Plains. *Remote Sensing of Environment* 108, 290–310. doi:10.1016/
677 j.rse.2006.11.021.

678 Yang, S., Gu, L., Li, X., Jiang, T., Ren, R., 2020. Crop classification method
679 based on optimal feature selection and hybrid CNN-RF networks for multi-
680 temporal remote sensing imagery. *Remote Sensing* 12, 3119. doi:10.3390/
681 rs12193119.

682 Zhu, X.X., Tuia, D., Mou, L., Xia, G.S., Zhang, L., Xu, F., Fraundorfer, F.,
683 2017. Deep learning in remote sensing: A comprehensive review and list of
684 resources. *IEEE Geoscience and Remote Sensing Magazine* 5, 8–36. doi:10.
685 1109/MGRS.2017.2762307.